Remaining Useful Life Prediction of Turbofan Engines Using Deep Learning

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I. Abstract

Predicting the **Remaining Useful Life (RUL)** of aircraft **turbofan engines** is a key task in predictive maintenance and prognostics health management (PHM). Accurate estimates help reduce costs, plan maintenance effectively, and avoid unexpected failures. In this work, we use the **NASA CMAPSS dataset**, which includes four subsets (FD001–FD004) with increasing complexity in terms of operating conditions and fault modes.

We employ a **Long Short-Term Memory (LSTM)** network to capture **temporal degradation patterns** in multivariate sensor data. To account for varying operating conditions, we apply condition-wise normalization and include the operating condition as an additional input feature. Feature selection is used to remove uninformative sensors, and a **sliding window** approach is adopted to generate input sequences.

Models are trained and evaluated separately on each dataset using **RMSE**, **MAE**, and $\mathbf{R^2}$. The results show that the LSTM performs well overall, though performance decreases for more complex datasets with multiple conditions and fault modes. These findings highlight both the effectiveness and the limitations of standard LSTM models for RUL prediction.

II. Introduction

II.1 Problem Statement

Estimating the **RUL** of turbofan engines from sensor data is not straightforward. Sensor readings are influenced by changing operating conditions, different fault types, and noise, which makes it hard to clearly separate actual degradation from other effects and reliably assess engine health.

II.2 Objectives

The objective of this work is to develop a **data-driven model** that can learn **temporal degradation patterns** from **multivariate time-series data** and provide accurate and generalizable RUL predictions across datasets with varying levels of complexity.

III. Data Status

III.1 Overview of CMAPSS Dataset

The **CMAPSS dataset** is a NASA benchmark for RUL prediction, containing simulated **run-to-failure data** of multiple turbofan engines. Each engine is represented as a **multivariate time series** of sensor readings and operating settings over time. The dataset is divided into four subsets with varying operating conditions and fault modes.

III.2 Dataset Variants

Each dataset contains multivariate time series split into training and test sets. Engines start with varying initial wear (considered normal), operate under three settings affecting performance, and the sensor data includes noise.

III.2.1 FD001

100 train / 100 test trajectories, 1 condition (sea level), 1 fault mode (**HPC degradation**)

III.2.2 FD002

260 train / 259 test trajectories, 6 conditions, 1 fault mode (**HPC degradation**)

III.2.3 FD003

100 train / 100 test trajectories, 1 condition (sea level), 2 fault modes (**HPC and fan degradation**)

III.2.4 FD004

248 train / 249 test trajectories, 6 conditions, 2 fault modes (**HPC and fan degradation**)

III.3 Data Format

Each engine is identified by a unit number and tracked over time cycles until failure, capturing its degradation over time. At each cycle, data is recorded as a combination of **operational settings** and **sensor measurements**.

The dataset includes three operational settings that define the operating condition, along with multiple sensors monitoring engine behavior. These readings form a **multivariate time series**, though some sensors may be noisy or uninformative.

IV. Methodology

IV.1 Overview of Pipeline

The proposed pipeline processes raw multivariate time-series sensor data through a structured sequence of stages to extract meaningful degradation patterns and predict RUL across all four CMAPSS sub-datasets. The key stages are summarised in Table 1.

Stage	Description
RUL Computation	Derive RUL from maximum cycle count per unit and apply piecewise clipping at 125 cycles
Feature Selection	Remove constant and uninformative sensors based on correlation with RUL
Signal Smoothing	Apply per-unit rolling mean to reduce sensor noise
Fault Mode Analysis	Identify implicit fault modes in FD003/FD004 via K-Means clustering and PCA validation
Normalisation	Apply global or condition-wise StandardScaler to remove operating point bias
Modelling	Train and compare Linear Regression, Random Forest, FNN, and CNN-LSTM models for RUL prediction

Table 1: Overview of the proposed pipeline

IV.2 Data Preprocessing

IV.2.1 RUL Computation

The **Remaining Useful Life (RUL)** for each engine is computed as the number of cycles remaining until failure. For the training data, which contains full run-to-failure trajectories, the RUL at any time step is given by:

$$\text{RUL} = \text{max_cycle}_{engine} - \text{current_cycle}$$

IV.2.2 Feature Selection

The CMAPSS dataset contains **21 sensors**, many of which are uninformative for RUL prediction due to near-constant behaviour. Including them adds noise and unnecessary complexity.

Feature selection is performed using visual inspection and absolute Pearson correlation:

$$\rho = \frac{|\text{Cov}(s_i, \text{RUL})|}{\sigma_{s_i} \cdot \sigma_{\text{RUL}}}$$

Sensors with low correlation and flat trends are removed. Accordingly, **sensors 1, 5, 6, 10, 16, 18, and 19** are discarded, retaining **14 informative sensors** across all datasets.

IV.2.3 ***Signal Smoothing***

To reduce cycle-to-cycle noise, a rolling mean is applied as a low-pass filter. A window size of $k = 5$ (vs. 3 and 7) provides the best balance between noise reduction and preserving degradation trends.

IV.2.4 ***Normalisation***

Sensor scales vary and shift across operating conditions. To address this, **condition-wise normalisation** is applied by fitting separate scalers per condition, ensuring the model learns degradation patterns rather than regime differences.

IV.3 Fault Mode Analysis

FD003 and FD004 contain two **fault modes**. These are identified using **K-Means clustering** ($K = 2$), and the resulting cluster assignments are used as fault mode labels for each engine.

IV.4 Predictive Modelling

Following preprocessing and fault mode analysis, a set of models with increasing complexity is trained to predict **Remaining Useful Life (RUL)**. These range from a simple linear baseline to deep learning models capable of capturing nonlinear and temporal relationships in the data.

All models are evaluated using a consistent prediction framework to ensure fair comparison across approaches.

IV.4.1 Linear Regression

A **Linear Regression** model was used as a baseline to establish a reference level of performance. The model is trained by minimising mean squared error over the training data. Although simple, it serves as a useful benchmark to assess whether more complex models capture additional structure in the data.

IV.4.2 Random Forest

A **Random Forest** regressor is used as a strong nonlinear baseline. The model consists of an ensemble of decision trees trained on bootstrapped samples of the data, with random feature selection at each split. The final prediction is obtained by averaging the outputs of all trees. In this work, $N = 100$ trees are used. Random Forest is effective at capturing nonlinear relationships and is relatively robust to noise, making it suitable for datasets with varying sensor behaviour.

IV.4.3 Feedforward Neural Network (FNN)

A **Feedforward Neural Network** is used to model nonlinear relationships between sensor features and RUL. The network consists of three hidden layers with 128, 64, and 32 units respectively.

Batch Normalisation is applied after each layer to stabilise training, and **Dropout** with rate $p = 0.3$ is used to reduce overfitting. The network is trained using the **Adam optimiser** with a learning rate of 0.001. The model is trained for 20 epochs with a batch size of 64, using mean squared error as the loss function. A validation split of 10% is used to monitor training performance.

IV.4.4 LSTM Model

To capture temporal behaviour, a **Long Short-Term Memory (LSTM)** network is used. It maintains a **cell state** $\mathbf{c}_t$ for long-term memory, controlled by forget, input, and output gates that regulate information flow.

Input data is structured using a **sliding window** of length W, forming sequences of shape (W, d). The LSTM processes these sequences and the final hidden state $\mathbf{h}_W$ is used to predict RUL:

$$\hat{\text{RUL}} = \mathbf{W}_{\text{out}}\mathbf{h}_W + b_{\text{out}}$$

The model is trained using **Adam** with MSE loss, and **dropout** is applied to improve generalisation.

V. Results

V.1 Feature Selection and Interpretation

Feature selection was performed in two steps. First, all 21 sensors were plotted against RUL for a sample of engine units to visually identify sensors showing no consistent degradation trend. Second, the absolute Pearson correlation between each sensor and RUL was computed.

Sensors 1, 5, 6, 10, 16, 18, and 19 were found to be nearly constant across all operating cycles, exhibiting both flat visual profiles and near-zero correlation with RUL. These were removed from all subsequent analysis.

The remaining 14 sensors all showed meaningful correlation with RUL, as shown in Figure 2. Among these, `sensor_11` showed the strongest correlation ($|\rho| = 0.696$), followed by `sensor_4` ($|\rho| = 0.679$) and `sensor_12` ($|\rho| = 0.672$). Sensors such as `sensor_14` and `sensor_9` showed weaker but still non-negligible correlations and were retained. This feature set was applied consistently across all four datasets.

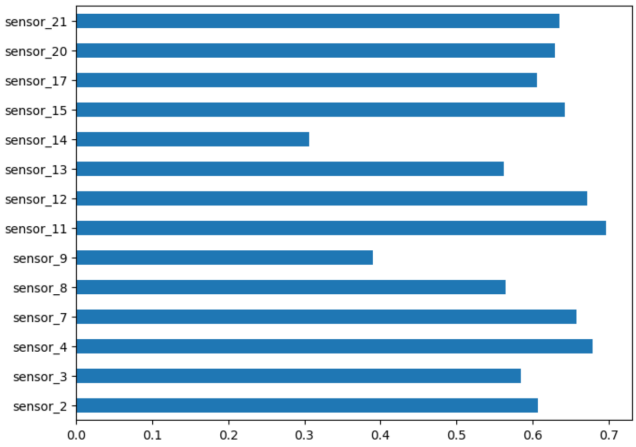


Figure 1: Absolute Pearson correlation of retained sensors with RUL (FD001).

V.2 Fault Mode Analysis

FD003 and FD004 contain two implicit fault modes. To identify these, each engine was represented by the mean of its final 30 cycles of sensor readings, which were then standardised and passed to a K-Means algorithm with K=2.

The clustering produced two groups: **44 engines in cluster 0** and **56 engines in cluster 1** for FD003. To validate that these clusters represent genuinely distinct degradation behaviours rather than arbitrary partitions.

As a further check, sensor trajectories were plotted over time coloured by cluster label. Distinct degradation patterns were visible across clusters, supporting the interpretation that the two groups correspond to the two underlying fault modes present in FD003 and FD004.

V.3 Results on datasets

V.3.1 FD001

FD001 serves as the baseline dataset, comprising a single operating condition and a single fault mode, making it the least complex of the four datasets. **Sensor correlation analysis** identified 14 informative sensors, with `sensor_11`, `sensor_4`, and `sensor_15` showing the strongest negative correlation with RUL, indicating consistent degradation signals.

Seven sensors (`sensor_1`, `sensor_5`, `sensor_6`, `sensor_10`, `sensor_16`, `sensor_18`, `sensor_19`) were

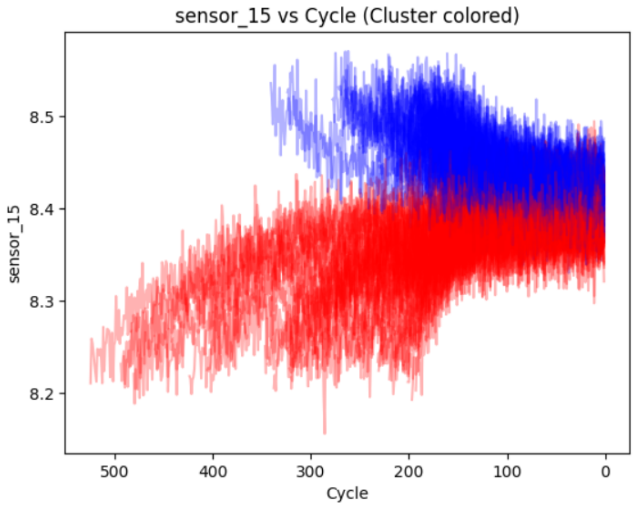


Figure 2: .Prediction of mode of failure from sensors data

dropped due to near-zero correlation. Table 2 summarises the performance of all models evaluated on this dataset.

Random Forest achieved the best test performance, though a notable gap between training RMSE (6.76) and test RMSE indicates **moderate overfitting**. Applying a sliding window of size 5 for smoothing did not improve **Linear Regression** performance, suggesting that sensor noise is already low under a single operating condition. The **FNN** achieved comparable performance to Random Forest, converging stably over 20 epochs.

Table 2: Model Performance on FD001

Model	**RMSE**	**MAE**	$\mathbf{R^2}$
Linear Regression	20.82	16.61	0.730
LR + Sliding Window	22.42	17.79	0.687
Random Forest	17.78	13.19	0.817
FNN	17.84	12.76	0.816

V.3.2 FD002

FD002 introduces six operating conditions while retaining a single fault mode, significantly increasing **dataset complexity**. The operating condition was encoded as an additional feature and included alongside sensor readings. Table 3 summarises model performance.

All models degraded relative to FD001, reflecting the added difficulty of **multi-condition prediction**. **Linear Regression** performed poorly, consistent with the nonlinear interaction between operating conditions and sensor degradation. **Random Forest** improved on this but still exhibited **heavy overfitting**, with a training RMSE of 7.29 versus a test RMSE of 29.12.

Table 3: Model Performance on FD002

Model	**RMSE**	**MAE**	$\mathbf{R^2}$
Linear Regression	32.64	24.34	0.632
Random Forest	29.12	20.25	0.707
FNN	31.70	22.86	0.652
FNN + Condition-Aware Scaling	29.58	20.41	0.697

The **FNN** with standard scaling scored lower than Random Forest. However, **condition-aware normalisation**, where each operating condition was scaled independently, improved FNN perfor-

mance to an RMSE of 29.58 and an R^2 of 0.697. This supports the hypothesis that consistent feature representation across conditions benefits **nonlinear models** more than linear ones.

V.3.3 FD003

FD003 shares the single operating condition of FD001 but introduces two **implicit fault modes**, adding a source of heterogeneity not directly observable in the sensor readings. To identify these fault modes, each engine was represented by the mean of its final 30 cycles of sensor readings. These features were then standardised and passed to a **K-Means** algorithm with $K = 2$. The clustering produced two groups: 44 engines in cluster 0 and 56 engines in cluster 1.

Table 4: Model Performance on FD003

Model	RMSE	MAE	R^2
Linear Regression	23.757	18.563	0.671
LR + Averaged Predictions	22.817	17.614	0.696
LR + Sliding Window (w=3)	18.344	14.054	0.796
Random Forest	21.240	16.066	0.737
FNN	21.017	15.837	0.742
LSTM	11.761	8.668	0.910

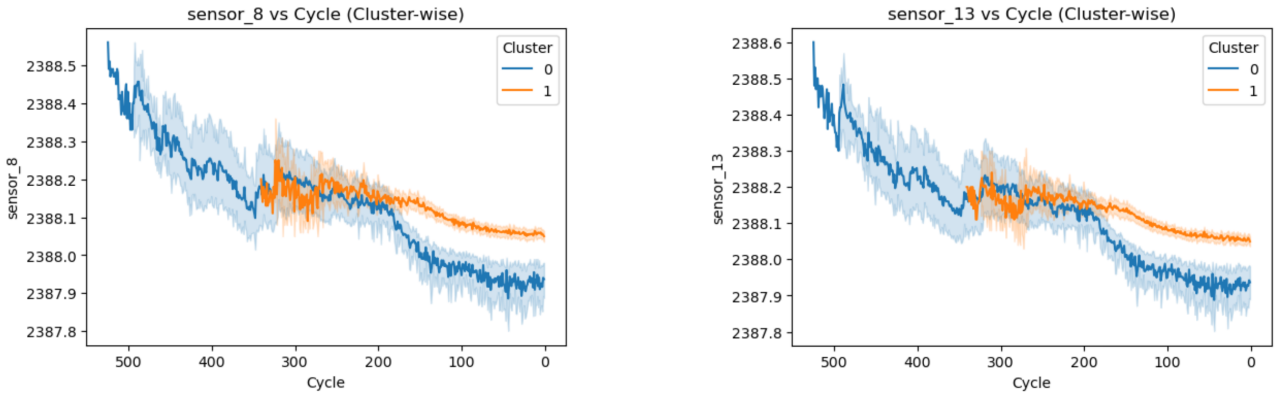


Figure 3: Mean Degradation Trajectory of sensor 8 and 13 by Fault Mode Cluster (FD003)

V.3.4 FD004

FD004 is the most complex dataset, combining six operating conditions with two implicit fault modes. Initial experiments with simpler models showed poor performance, consistent with findings on FD002, where increased operating condition complexity degraded all baseline models.

Two architectures were evaluated: a standard LSTM and a Bidirectional LSTM (**BiLSTM**), both trained on sliding window sequences of length 30 across 15 selected sensors. Table 5 summarises model performance.

The BiLSTM outperformed the standard LSTM across all metrics, achieving an RMSE of 16.64, MAE of 10.78, and an R^2 of 0.850. The **NASA asymmetric score**, defined as

$$S = \sum_{i=1}^{N} \begin{cases} e^{-\frac{e_i}{13}} - 1, & e_i < 0 \\ e^{\frac{e_i}{10}} - 1, & e_i \geq 0 \end{cases}$$

where $e_i = \hat{y}_i - y_i$, penalises late predictions more heavily than early ones. It further confirms the BiLSTM's superiority, with a score of 5,060 compared to 6,812 for the standard LSTM.

V.4 Cross dataset comparison

Across all four datasets, model performance degraded predictably as dataset complexity increased, validating the **progressive difficulty structure** of the CMAPSS benchmark.

Table 5: Model Performance on FD004

Model	RMSE	MAE	R^2
LSTM	17.51	11.63	0.834
BiLSTM	16.64	10.78	0.850

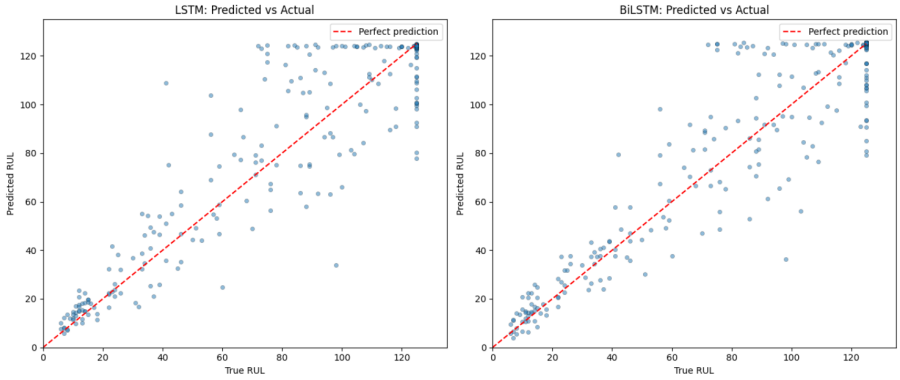


Figure 4: Predicted vs Actual RUL for LSTM and BiLSTM on FD004

FD001, with a single operating condition and a single fault mode, yielded the strongest baseline results. Both Random Forest and FNN achieved an R^2 of approximately 0.817.

FD002, which introduces six operating conditions, showed a significant drop in performance across all models. The best test RMSE increased from 17.78 to 29.12, highlighting the challenge of **multi-condition degradation prediction** for models that do not explicitly account for operating regime.

FD003, despite sharing FD001's single operating condition, proved more difficult due to the presence of two **implicit fault modes**. Identifying these fault modes via K-Means clustering and treating them as a source of heterogeneity provided important context for interpreting sensor trajectories. However, the direct impact on model performance was limited, since cluster labels were not used as explicit features.

FD004, which combines both sources of complexity, was the most challenging dataset and motivated the use of **sequential models**. The BiLSTM achieved the best overall performance across all datasets, with an RMSE of 16.64 and an R^2 of 0.850. This demonstrates that **temporal modelling** of sensor sequences is particularly effective when both operating condition variability and fault mode heterogeneity are present.

Table 6: Best Model Performance Across All Datasets

Dataset	Best Model	RMSE	MAE	R^2
FD001	Random Forest	17.78	13.19	0.817
FD002	Random Forest	29.12	20.25	0.707
FD003	LSTM	11.761	8.668	0.910
FD004	BiLSTM	16.64	10.78	0.850

VI. Discussion

Results across the CMAPSS datasets show a clear link between **dataset complexity** and performance. FD001 provides a strong baseline where simple models perform well under a single condition. Performance drops on FD002 due to multiple operating conditions, highlighting the importance of **feature representation** in multi-condition settings. FD003 performs better as it retains a single condition, allowing models to focus on degradation dynamics.

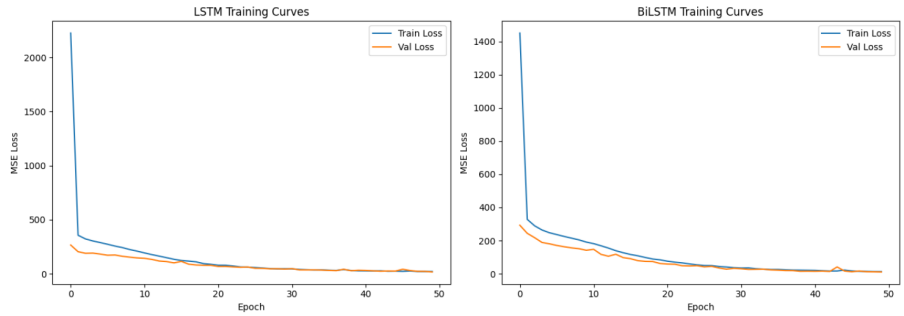


Figure 5: Loss function comparison for LSTM and BiLSTM on FD004

VI.1 Temporal Modelling

LSTM achieves the best result on FD003 ($R^2 = 0.910$), demonstrating the effectiveness of **temporal modelling** under stable conditions.

VI.2 Generalisation and Limitations

Random Forest shows consistent overfitting, indicating difficulty handling **distribution shifts** across engines.Limited tuning and preprocessing differences may affect results.

VI.3 Fault Structure

Clustering reveals meaningful **fault modes**, suggesting potential gains from incorporating these labels into models.

VII. Conclusion

This study evaluated RUL prediction on CMAPSS using models from Linear Regression to BiLSTM.

FD001 establishes a strong baseline with simple models. FD002 highlights the role of preprocessing under multiple conditions. FD003 shows the strength of LSTM ($R^2 = 0.910$), while FD004 confirms the advantage of **BiLSTM** (RMSE 16.64, $R^2 = 0.850$, lower NASA score).

Clustering reveals interpretable fault modes, offering scope for future model conditioning. Overall, model choice should match dataset complexity, with sequential models preferred for FD003 and FD004.

VIII. References

1. Christopher Olah (2015). *Understanding LSTM Networks*. Available at: `https://colah.github.io/posts/2015-08-Understanding-LSTMs/`
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3. Egehanyorulmaz (n.d.). *NASA Turbofan Engine RUL Prediction*. GitHub repository. `https://github.com/egehanyorulmaz/nasa-turbofan-engine-rul-prediction`